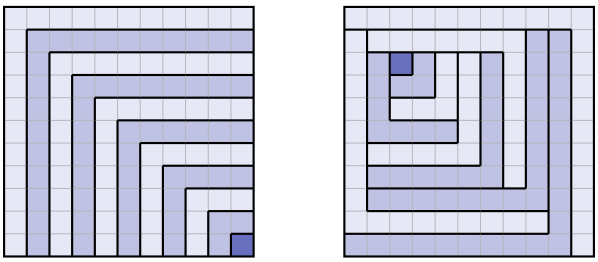
BMO1 2025 (November 2025), Problem 3

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The problem

An 11×11 *square grid is dissected into* 11 *pieces as follows. At each step, remove the top row and the leftmost column of the remaining grid as a single piece. Thus, the pieces obtained are L-shaped with sizes* $21, 19, \ldots, 3$*, together with a single square. The pieces are rearranged to reassemble an* 11×11 *grid, with rotations allowed (but the pieces may not be turned over) so that the single square occupies the cell in the third row and third column (counting from the top left). How many ways can this be done?*



the eleven pieces one reassembly

The answer is $\binom{10}{2}^2 = 45^2 = 2025$. Call the piece of size $2k-1$ the *k-piece*. Each arm of the k-piece is k cells long, counting the corner cell in both arms. On the right, the 11-piece and the 7-piece take a top row and the 9-piece and the 4-piece take a left column, which puts the single square in row 3, column 3.

Script A **10/10**

Call the piece of size $2k-1$ *the k-piece. In every rotation the k-piece has a horizontal arm and a vertical arm of k cells sharing a corner cell.*

Step 1 (nesting). *We claim that for* $k = 11, 10, \ldots, 2$ *the pieces of sizes* $2k-1, \ldots, 3$ *and the single square fill a* $k \times k$ *square* S_k*, with the k-piece filling a boundary row and a boundary column of* S_k*. Induct downwards, starting from* S_{11}*, the whole grid. The horizontal arm of the k-piece has k cells inside* S_k*, which is k wide, so the arm fills a row of* S_k *and the corner cell is at an end of that row. Hence the vertical arm lies in the leftmost or rightmost column of* S_k*. Similarly the vertical arm fills that column, so the horizontal arm is the top or bottom row. Removing this row and column leaves a* $(k-1) \times (k-1)$ *square* S_{k-1}*, filled exactly by the remaining pieces.*

Step 2 (independence). *So an arrangement is a list of choices, top or bottom and left or right, for* $k = 11, \ldots, 2$*. Each of the four combinations fixes the rotation of the k-piece (the four rotations put the corner at the four corners). Every list gives an arrangement, and different lists give different arrangements, since at the first difference the same piece is placed differently. The top/bottom choices decide only rows and the left/right choices only columns, so the count is (number of vertical lists)×(number of horizontal lists).*

Step 3 (counting). *Each top choice moves the top row of the remaining square down by one and each bottom choice does not, so* S_1 *is in row* $1 + t$*, where t is the number of tops. We need* $t = 2$*, which gives* $\binom{10}{2} = 45$ *vertical lists. Similarly there are* 45 *horizontal lists with exactly two lefts.*

Hence the answer is $45^2 = 2025$*.*

Marker's comment. Complete. Step 1 proves the nesting rather than assuming the pieces go back the way they came apart, which is the point the markers' report singles out. Step 2 says why the vertical and horizontal choices combine freely and why different lists give different arrangements, which multiplying two binomial coefficients silently skips.

Script B **8/10**

Let the k-piece be the piece of size $2k-1$, with two arms of length k. The 11-piece has an arm of length 11 in an 11×11 grid, so this arm is a whole row, and the other arm is a whole column. Since the arms meet at an end of each, the row is the top or bottom row and the column is the leftmost or rightmost column. The rest of the grid is a 10×10 square filled by the other pieces, and repeating the argument, each k-piece fills a boundary row and a boundary column of a $k \times k$ square.

The single square ends in row 3 when exactly two of the ten pieces take the top row of their square, which can happen in $\binom{10}{2} = 45$ ways. Similarly the single square is in column 3 in $\binom{10}{2} = 45$ ways. So the answer is $45^2 = 2025$.

Marker's comment. The nesting, the expensive part, is proved. The count multiplies the row and column counts without saying why the choices are independent or why each choice of corner fixes the rotation, and the report records a small penalty for this. One sentence, as in Step 2 of Script A, earns the last marks.

Script C **3/10**

The grid was made by peeling off L-shapes from a corner, so to reassemble the grid the pieces must go back in the same nested way. The 11-piece goes round a corner of the grid, the 10-piece goes round a corner of the 10×10 square that remains, and so on down to the 2-piece in a 2×2 square, leaving the single square.

Each piece chooses top or bottom and left or right. The single square is in row 3 exactly when 2 of the 10 pieces choose top, and in column 3 exactly when 2 choose left. The vertical and horizontal choices are independent, so the answer is $\binom{10}{2} \times \binom{10}{2} = 2025$.

Marker's comment. The counting is right, but the nesting is assumed, and whether the pieces can fit together differently is what the question tests. The report says scripts without this step were heavily penalised. The missing argument is short. An arm of 11 cells in an 11×11 grid fills a row, so the corner of the 11-piece is at a corner of the grid, and induction does the rest.

Script D **1/10**

Try smaller grids, cut the same way, with the single square in the third row and third column. For a 3×3 grid there is 1 way, for a 4×4 grid there are 9 ways, for a 5×5 grid there are 36 ways and for a 6×6 grid there are 100 ways. These are 1^2, 3^2, 6^2, 10^2, squares of triangular numbers, so for an $n \times n$ grid the answer is $\binom{n-1}{2}^2$. For $n = 11$ the answer is $\binom{10}{2}^2 = 2025$.

Marker's comment. The small values and the answer are right, but the formula is a guess from a table and even the small values come without justification. One mark for the correct answer with sensible exploration.

What full marks needs

Prove that every arrangement is nested, using the fact that an arm of length k fills a whole row or column of a $k \times k$ square. Then count, saying why the vertical and horizontal choices are independent and why each list needs exactly two tops or two lefts.